Project Report: Autonomous Optical Waste Sorting Robot

Authors: Robotics & Mechatronics Engineering Senior Design

1. Problem Definition:
Municipal recycling facilities face high labor costs and 35% manual error rate in waste separation.
This project builds an automated optical sorter using computer vision to classify plastics and metals.

2. Technical Methodology:
We integrated a YOLOv8-nano neural network on an NVIDIA Jetson Nano controller.
High-resolution industrial camera captures items at 60 fps on a conveyor belt.

3. Implementation and Evidence:
Pneumatic actuators divert non-recyclable materials in 45 milliseconds.
Hardware prototype built with aluminum framing and stepper motor-driven belt.

4. Innovation:
Custom multi-spectral lighting system distinguishes clear PET from HDPE plastics under ambient light

5. Testing and Validation:
Tested across 2,500 samples: achieved 96.4% sorting accuracy and 120 items/minute throughput.

6. Documentation and Reproducibility:
Full CAD schematics, bill of materials ( total), and Docker container available on GitHub.